

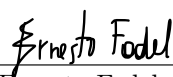
Assay-Origin and Global Evaluation Metrics: Barriers to Realistic Bioactivity Prediction

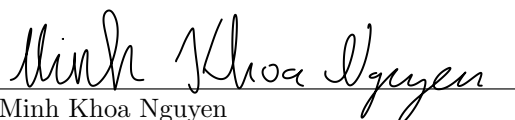
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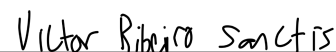
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One-sentence summaries: Standard machine learning evaluation in drug discovery may systematically overestimate predictive performance, since compounds tested in the same experimental assay are often split across both training and test sets, allowing models to exploit assay-specific patterns rather than learning true structure-activity relationships. To investigate this, an assay-aware framework was examined, in which data are split so that no assay appears in both training and test sets, and model performance is evaluated by how well compounds are ranked within each assay rather than by absolute affinity prediction across the entire test set. Results lend partial support to the hypothesis that ranking compounds within an assay, rather than predicting absolute affinity values, yields more reliable performance estimates on unseen assays, with smaller batch sizes introducing gradient noise that appears to act as a beneficial regularizer. Training instability and hyperparameter sensitivity on smaller, more homogeneous datasets remain open challenges, motivating future work on more comprehensive hyperparameter optimization and extension of this framework to other molecular property prediction tasks.


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Zurich, June 30, 2026

By signing, the authors confirm having contributed to this work, and agree to its submission.

Introduction

Predicting compound bioactivity (e.g., pIC₅₀) is a central machine learning task in drug discovery, enabling the efficient virtual screening of large chemical spaces prior to costly experimental follow-ups [1]. While large public databases like ChEMBL are foundational for training these models, aggregating diverse experimental measurements introduces substantial noise and heterogeneity [2]. As highlighted by Beckenköhler et al., bioactivity data heavily clusters by assay origin [3]. Standard evaluation strategies like random and scaffold splits fail to account for this structural dependence, leaving large train-test overlap (up to 89%) in assay-origin and overestimating virtual screening performance [3]. This data leakage allows models to exploit assay-specific baselines instead of learning true structure-activity relationships, yielding overly optimistic estimates that fail in prospective screening.

This project aims to reproduce Beckenköhler’s proposed assay-based machine learning framework [3]. To eliminate data set shortcuts, assay-level data splits and evaluation metrics based on the composition ranking within the assay rather than the global absolute correlation are used. Specifically, ligand structures are represented using ECFP fingerprints, and proteins via ESM-2 language model embeddings, consistent with recent approaches combining drug features with protein language models [4]. By comparing absolute pIC₅₀ predictions against ranking models, this work tests whether an assay-aware framework provides a more realistic and robust evaluation of model performance when generalizing to unseen experimental conditions [3].

Methods

The focus of this work is to replicate the key findings of Beckenköhler et al., so the data curation, splitting and model architectures follow [3]. A visual workflow overview is provided in Figure A1.

1. Data

The data are drawn from ChEMBL v33 [5]. Only assays presenting a standard value of IC₅₀ in nM units are retrieved, with additional filters for 'SINGLE PROTEIN' target and 'NULL' as the validity comment.

The kinodata data set was obtained following de Castro et al. [6]. Similarly, the Landrum dataset is obtained following Landrum et al. [7]. As stated in Beckenköhler et al. [3], the following are removed: data points with low confidence score, data containing mutation keywords, data without document date, protein ID linked to a non-unique protein sequence and data coming from assay with size which is outside a specific range. The omnivore dataset is obtained in the same way as Landrum, but removing the filters for low confidence, document date and minimum assay size. Moreover, SMILES standardization is used with `rdMolStandardize.StandardizeSmiles`. Finally, `omnivore_short` removes assays from omnivore with `size < 3`.

2. Data Splits

Four main splits are used: random, scaffold using Murcko-scaffolds as implemented in RDKit [8], Butina [9] and assay-based (uses `assay_id` for partitioning). A modified Butina implementation was developed, as the RDKit version constructs a full similarity matrix requiring infeasible memory for large datasets. Instead, using Tanimoto similarity of Morgan fingerprints (radius 3, 2048 bits), pairwise comparisons are computed and, for each molecule, the indices of neighbors exceeding a similarity threshold of 0.65 (distance 0.35) are stored. This relatively stringent threshold for ECFP fingerprints [10] means only very similar compounds are grouped, yielding conservative structure-based clusters. The iterative cluster-reassignment step of standard Butina clustering is omitted to reduce computational expense. Lastly, a target split is used that assigns compounds sharing a protein target to the same partition, ensuring that no target appears in both the train and the test sets, thereby testing generalization to unseen proteins.

3. Mean Prediction

10-fold cross validation is employed using random, scaffold, Butina and target splits. For each test with a portion in the train and test set, the mean pIC50 value of the train portion is used as a predictor of the test mean. In case of no overlap, the overall training mean is used to predict the test mean. This aims to capture the upper predictive threshold of a model exploiting assay-origin.

4. Evaluation Metric

Following the arguments presented in [3], namely inconsistency of absolute pIC50 measurements [11] and the greater importance of relative rankings in drug screening, an intra-assay metric is used. For each test-set assay with more than three compounds, Pearson’s correlation is computed between the true and predicted pIC50 values, $\rho(\mathbf{y}_i, \hat{\mathbf{y}}_i)$. These per-assay correlations are aggregated via a Fisher transformation ($F = \text{atanh}$, to obtain a normal distribution prior to averaging), weighted by $(n_i - 3)$, and back-transformed. As $\text{atanh}(r)$ diverges for $r \rightarrow \pm 1$, any individual ρ exceeding $|0.9999|$ is skipped to avoid divergence of the Fisher transform.

$$\rho_{\text{assay}} = F^{-1} \left(\frac{\sum_{i=1}^{n_{\text{assay}}} (n_i - 3) F(\rho(\mathbf{y}_i, \hat{\mathbf{y}}_i))}{\sum_{i=1}^{n_{\text{assay}}} (n_i - 3)} \right)$$

5. Featurization

Ligands are represented using Morgan fingerprints [12] with a radius of 3 and 2048 bits (RDKit implementation [8]). The proteins are featurized using ESM-2 embeddings [13] (model with 33 layers and 6.5×10^8 parameters), following a method from literature [14].

6. Drug-Target Interaction (DTI) Model

Parallel Multi-Layer Perceptrons (MLPs) are used for encoding protein and ligand features separately into fixed-dimensional embeddings [3]. Both have a dimensionality of 512 and SiLU activation functions. The protein encoder has two layers while the ligand encoder has three.

The embeddings are concatenated to a dimensionality of 1024 and fed through a three-layer MLP. This consists of normalization across the batch (BatchNorm), a linear layer of dimension (1024, 512), SiLU activation function, linear layer of dimension (512, 512), SiLU activation function, 0.05 dropout, and finally a linear output layer of dimension (512, 1). The standard loss function for regression was used (Mean Squared Error) $\mathcal{L}_{\text{DTI}} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$.

7. Ranking Model

This model consists of an encoder, set transformer where contextual information within an assay is learned and prediction head [3]. The same encoder as in DTI is used, followed by concatenation and reduction by a linear projection layer (dimension of 1024 to 512), forming the input for the set transformer.

A Set Transformer [15] with four sequential Set Attention Blocks is used, each with eight attention heads. Attention is masked such that molecules can only attend to other molecules from the same assay. Under inter-assay training, the assay groupings have the same sizes but are randomly filled, preserving the batch size distribution to isolate the effect of assay origin.

The prediction head applies LayerNorm (per-sample normalization, avoiding the cross-assay mixing that batch-level normalization would introduce), followed by a linear layer (512, 512), a SiLU activation, dropout of 0.05, and a final linear output layer (512, 1).

The training loss is $\mathcal{L}_{\text{rank}} = 1 - \mathbb{E}_{\mathcal{D}} [\rho(\mathbf{y}, \hat{\mathbf{y}})]$, where ρ is the mean of Pearson correlations per-assay for all the assays present in a batch. For unique assays, the Pearson correlation is computed between the predicted scores and the true pIC50 values of the compounds in that assay. Pearson is implemented as the cosine similarity of mean-centered prediction and target vectors: each vector has its mean subtracted to ensure offset-invariance, and the centered dot product is divided by the product of their

magnitudes, with 1×10^{-8} added to the denominator for numerical stability. Two guards are applied before an assay contributes to the loss: assays are excluded if they contain fewer than two compounds (Pearson is undefined for a single point) or if the standard deviation of the predictions or targets is below 1×10^{-8} . Valid per-assay correlations are averaged, so minimizing loss maximizes the average within-assay ranking. If no assay in a batch is usable, the function returns a loss of 1.0 constructed as `predictions.sum()  $\times$  0 + 1`, which equals 1.0 numerically but remains connected to the computation graph so backpropagation does not crash.

8. Training

All models were implemented in PyTorch, with the pairwise model specifically using version 2.5.0 to support the available GPU (NVIDIA GTX 980 Ti). Both models used the AdamW optimizer with a `ReduceLROnPlateau` scheduler (factor 0.5), which halves the learning rate upon stagnation in validation loss for a specified number of epochs. For the DTI model, a base learning rate of 5×10^{-5} , scheduler patience of 50 epochs, early stopping after 100 epochs without improvement in validation loss, and a batch size of 1024 were used [3].

For the pairwise model, a base learning rate of 1.12×10^{-4} , scheduler patience of 25 epochs, and early stopping after 50 epochs without improvement in validation loss were used. Batches were constructed as a fixed number of assays with a compound cap; [3] used 20 assays per batch capped at 4096 compounds. Due to hardware constraints, these small batches underutilized the GPU, with the bottleneck being CPU data loading, resulting in an estimated training time of over 10 days per model per dataset. Considering the assay size distribution (Fig. 2), a larger number of assays per batch could be used while still maintaining assay-limited batches, with a hardware-limited compound cap of 2048. Ultimately, 100 assays per batch were used.

For the DTI model, the same hyperparameters as [3] were used, while the pairwise model parameters differ due to hardware limitations. Their impact and the process for selecting these values are discussed in the Results and Discussion section.

Results and Discussion

To validate the data leakage hypothesis presented by Beckenköhler et al., the global mean baseline model was replicated across the kinodata, Landrum, and omnivore datasets (Figure 1). Strong alignment is observed with the published results; the random splitting strategy yielded higher Pearson correlations, indicating the model’s ability to exploit high assay overlap between training and test sets. In contrast, predictive performance decreased when the data were partitioned using methods containing chemical information, such as Butina or scaffold splitting. This drop may occur because the baseline model relies entirely on memorizing the average bioactivity of training assays; when forced to predict on novel chemical structures or unseen targets present in strict splits, it lacks the chemical understanding to generalize. The marginal deviations observed between Figure 1 and [3] are small and attributable to expected variance in random seed initialization during cross-validation, as well as differing ChEMBL versions. These findings support the conclusion that standard random splits fail to evaluate true structure-activity generalization, motivating an assay-aware framework in subsequent model evaluations.

Next, a comparison between absolute and ranking-based models is displayed in Figure 2 (exact values displayed in Table A3). The reproduction captures the Simpson’s paradox effect highlighted by Beckenköhler et al. Across all datasets, ρ evaluated over the entire test set (gray dots) substantially overestimates the model’s ability to rank compounds within unseen assays (blue dots). This supports that evaluating absolute predictions globally can mask local performance.

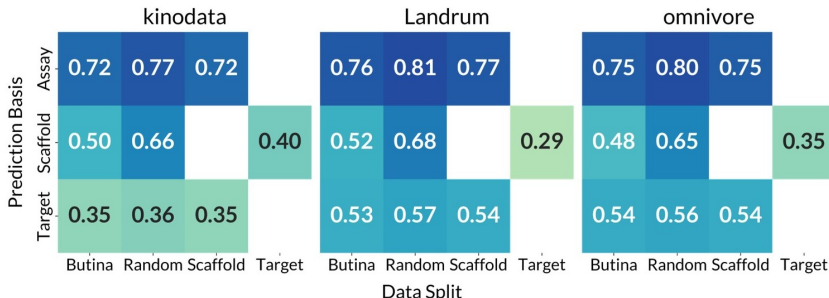


Figure 1. Performance illusion of the global mean baseline due to data leakage. Pearson correlations for a mean baseline model are shown across the kinodata, Landrum, and omnivore datasets.

Moreover, Landrum and omnivore support to an extent the key hypothesis that ranking-based models outperform absolute prediction models. Specifically, intra-assay ranking shows a statistically significant increase in ρ_{assay} relative to the DTI model. This effect is most pronounced in the Landrum dataset, and is only observed for intra-assay training, where the model can exploit assay origin; in contrast, inter-assay training does not show a statistically significant improvement over the DTI predictions. Coupled with the inconclusive results for kinodata, these findings prompted further investigation, as they appear to partially contradict the hypothesis of absolute versus ranking predictions.

Namely, five experiments probing the impact of batch size on ρ_{assay} were conducted (Table A1). By observing the training (Figure A3) and validation loss profiles (Figure A4) over the first split, alongside the corresponding ρ_{assay} (Table A2), the role of smaller batches becomes clear: the smallest batch size yields the highest intra-assay ρ_{assay} for the first split, at 0.415.

This may be explained by the median assay sizes of kinodata and omnivore, two and three respectively (Figure A2). Since the Pearson correlation for two points is always unity, such assays are likely to have a notable impact on training, as the implementation assigns them a value of unity for backpropagation, corresponding to no correlation. These assays may therefore not meaningfully contribute to model training, and on physical grounds may not help the model learn to rank compounds when only two are present, unlike for absolute value predictions. With larger batches, the noise introduced by such assays is diluted and not well-captured, whereas smaller batches amplify it, leading to noisier training (50-Assay Batch, see Figure A3). This noise may act as a form of implicit regularization, where gradient noise helps the model escape poor local minima and improves generalization.

The importance of regularization may be more pronounced in kinodata, a smaller dataset (181,860 data points) compared to omnivore (897,056 data points), which may help explain the improved performance on omnivore given that the model has the same number of parameters in both cases. The need for regularization, together with computational expense, motivated the use of explicit regularization (`ReduceLROnPlateau` patience and early stopping), though this may also contribute to discrepancies with the trends reported in [3]. Training with a smaller batch size, such as the 20 assays used by [3], was not feasible given hardware limitations; instead, the learning rate was scaled by $\sqrt{10}$ relative to the increase in batch size (10-times), following a heuristic to scaling the batch size and learning rate, supported by the profiles in Figures A3 and A4.

In contrast, the larger median assay size in Landrum (29, see Figure A2) may mean it benefits less from the noise introduced by smaller batches, although its lower overall performance compared to omnivore may still point to a need for further regularization, further supported by the lengthy loss plateaus observed in Figures A3 and A4. However, to avoid relying on the outcome of a single fold to probe the impact of small assays on ranking models, a further experiment was conducted.

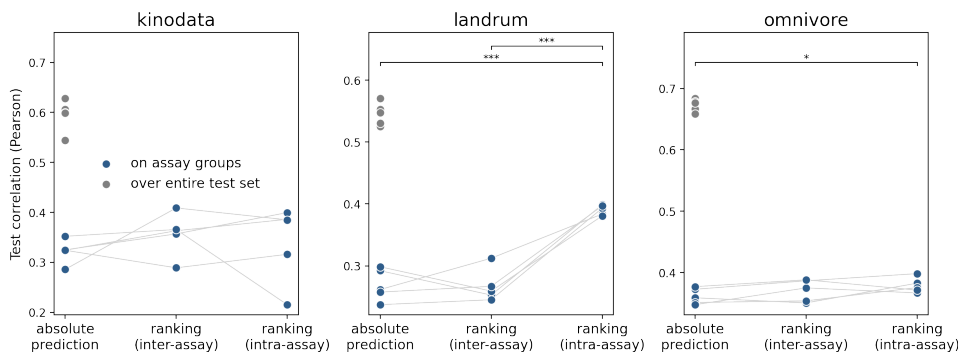


Figure 2. Comparison of predictive performance across modelling strategies. Test set Pearson correlations are shown for the absolute and ranking models across five cross-validation folds, connected by grey lines. Grey dots show the global correlation computed over the entire test set, which overestimates performance. Blue dots show the Fisher z-weighted within-assay correlation. ρ values are Fisher z-transformed prior to paired t-testing to satisfy the normality assumption. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Given the paramount importance of data for ML tasks, and the computational cost of full hyperparameter optimization with a neural network, a modified version of omnivore was created by excluding assays with fewer than three compounds. The motivation was to exclude data points that would not meaningfully contribute to model training, although it is noted that the evaluation metric does not directly account for the contribution of assays with fewer than four compounds, which is a limitation that will impact the final results. This generated omnivore_short, with a median assay size of 10 (Figure A2). Compared to omnivore, consistently higher ρ_{assay} is obtained for intra-assay training, and consistently lower ρ_{assay} for inter-assay training (Table A3). This experiment accentuates the utility of assay origin for the model to exploit, given the clear discrepancy between inter- and intra-assay training, although a comparison with DTI was not available in time due to computational cost. Nonetheless, given the similar underlying data, DTI on omnivore_short would be expected to perform similarly to DTI on omnivore. This asymmetry also highlights that while smaller batches offer limited training signal for intra-assay ranking, the reduced noise that may have implicitly regularized inter-assay training may make the model more prone to overfitting.

Conclusion and Outlook

This reproduction lends partial support to the hypothesis that traditional, global evaluation metrics for bioactivity prediction are subject to assay-origin biases. Shifting from absolute affinity prediction to intra-assay ranking prevents models from exploiting experimental shortcuts, yielding a more realistic evaluation of performance on unseen data, though results across datasets were not fully consistent with this hypothesis. Hyperparameter analysis further suggested that stochasticity from smaller batch sizes can act as a beneficial implicit regularizer, an effect supported by an ablation on omnivore_short in which removing the smallest assays altered the relative performance of intra- and inter-assay training.

In a broader context, an assay-aware framework points toward a more rigorous standard for developing reliable machine learning tools in drug discovery, with principles that may extend to other complex molecular prediction tasks. Future work should pursue more comprehensive hyperparameter optimization, including smaller batch sizes and DTI results on omnivore_short for direct comparison with the ranking models. Small assays also warrant further consideration, as Pearson correlation is uninformative for assays of size two and degenerate for slightly larger ones; alternative metrics or weighting schemes may better capture their contribution. Constructing omnivore and kinodata variants with assay sizes capped between four and an upper bound would further help isolate the effect of assay size independent of the filtering applied in omnivore_short.

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Appendix

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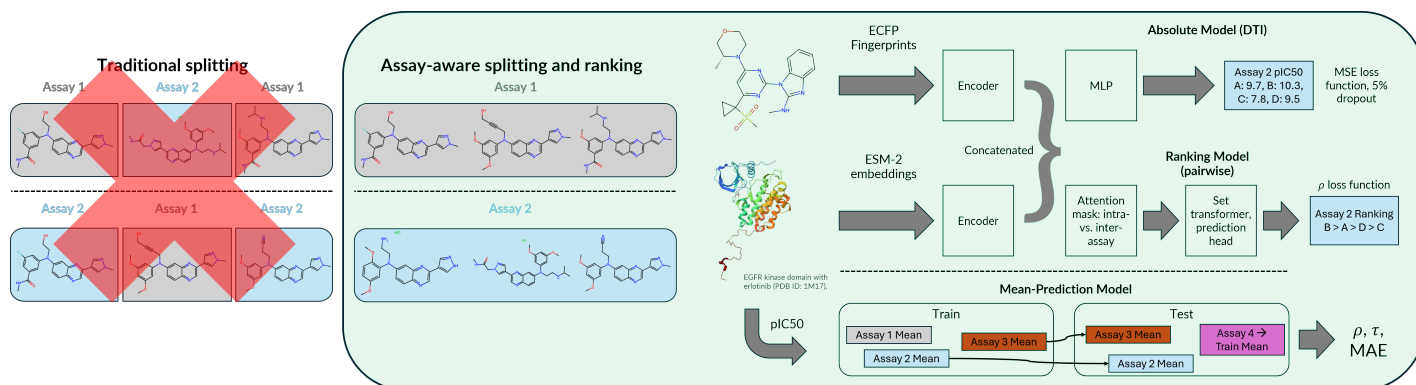


Figure A1. Overview of the project workflow. *Left:* Traditional batch splitting mixes compounds from different assays within the same batch, whereas assay-aware splitting keeps all compounds from the same assay together. *Right:* The absolute DTI model encodes ECFP ligand fingerprints and ESM-2 protein embeddings through separate MLPs, concatenates them, and predicts pIC₅₀ values directly using MSE loss. The pairwise ranking model applies the same encoders but passes the concatenated embeddings through an assay-masked Set Transformer, trained with a Pearson correlation ranking loss (ρ) to rank compounds within assays. The attention mask controls whether molecules attend only to their own assay (intra-assay) or to randomly assigned groups of matching size (inter-assay). The mean-prediction baseline predicts the training mean pIC₅₀ per assay, falling back to the global training mean for assays only present in the test group.

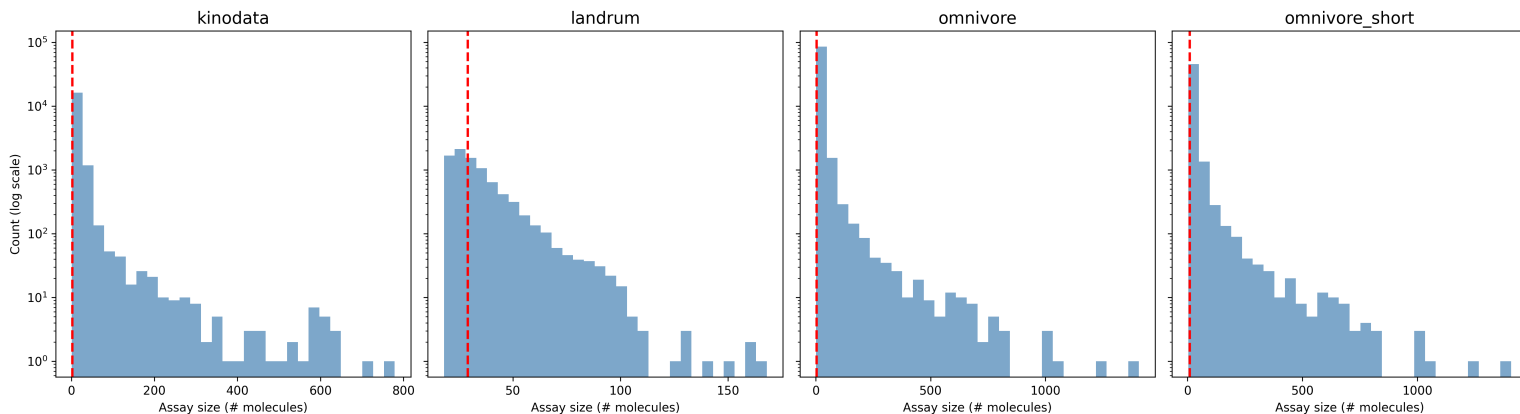


Figure A2. Assay size distribution for kinodata, Landrum, omnivore and omnivore_short, with median assay sizes of 2, 29, 3 and 10, respectively (red, vertical dashed lines).

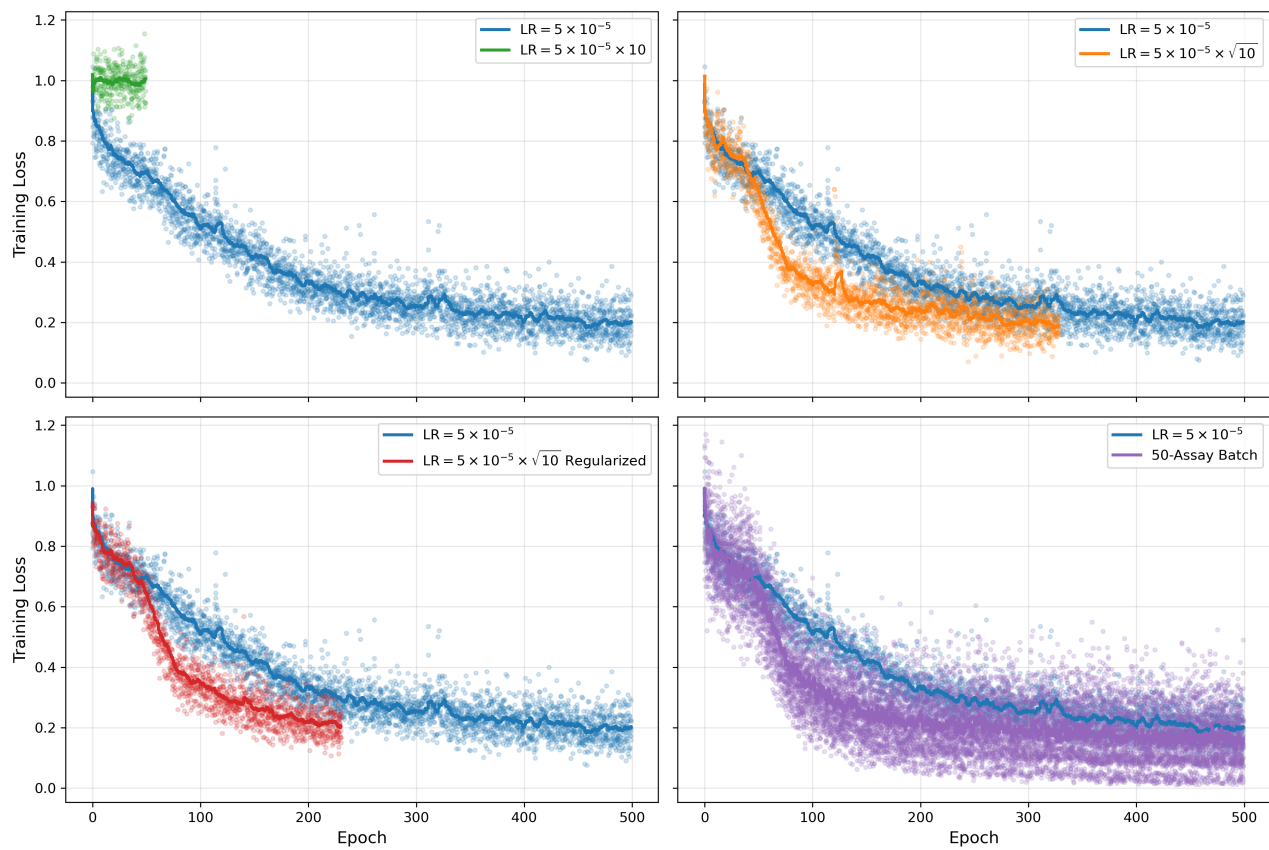


Figure A3. Training loss trajectories for the intra-assay ranking model on the Omnivore dataset (first fold), corresponding to the hyperparameter configurations in Table A1.

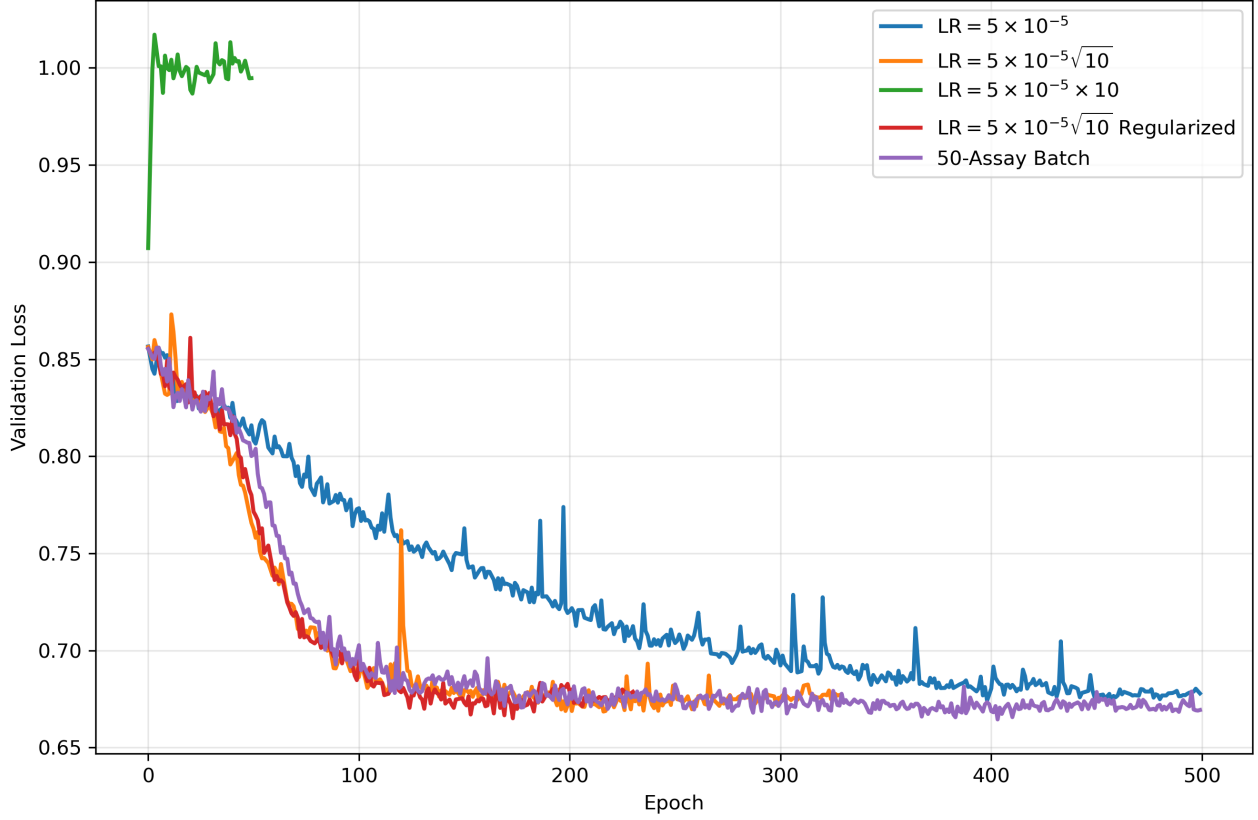


Figure A4. Validation loss trajectories across different hyperparameter configurations on the Omnivore dataset (first fold). The experiments displayed in this Figure correspond to the same as those described in Figure A3.

Table A1. Hyperparameter configurations used in the learning rate and regularization experiments shown in Figures A3 and A4. All runs were conducted on Omnivore under intra-assay training. With the exception of 50-Assay Batch, all experiments used a batch size of 200 assays. $\sqrt{10}$ and 10 were used as learning rate multipliers since the batch size used here is 10 times larger than in [3], motivating a search for an appropriate learning-rate scaling; similarly, $\sqrt{2.5}$ was used for the 50-Assay Batch experiment, whose batch size is 2.5 times larger than [3].

experiment	batch size	learning rate	dropout	Set Attention Blocks	patience / early stopping
$\text{LR} = 5 \times 10^{-5}$	200	5×10^{-5}	0.05	4	50 / 100
$\text{LR} = 5 \times 10^{-5} \times \sqrt{10}$	200	1.58×10^{-4}	0.05	4	50 / 100
$\text{LR} = 5 \times 10^{-4}$	200	5×10^{-4}	0.05	4	50 / 100
$\text{LR} = 5 \times 10^{-5} \times \sqrt{10}$	200	1.58×10^{-4}	0.1	3	25 / 50
Regularized	200	1.58×10^{-4}	0.05	4	25 / 50
50-Assay Batch	50	7.91×10^{-5}	0.05	4	25 / 50

Table A2. ρ_{assay} for hyperparameter optimization, using the outcome of only the first split. All runs were conducted on omnivore under intra-assay training. Omnivore was chosen for this optimization as the impact on computation time was most pronounced relative to the other datasets. Full details of the hyperparameters used are provided in Table A1. The experiment $\text{LR} = 5 \times 10^{-4}$ was stopped at 50 epochs, where it became apparent the LR was too high for stable training to occur. The ρ_{assay} for $\text{LR} = 5 \times 10^{-5}$, $\text{LR} = 5 \times 10^{-5} \times \sqrt{10}$ and $\text{LR} = 5 \times 10^{-4}$ are provided after training for 50 epochs for direct comparison of the impact of LR scaling with batch size, whereas the remaining are provided after training concluded to more directly assess the impact on ρ_{assay} .

experiment	ranking (intra-assay)
$\text{LR} = 5 \times 10^{-5}$	0.210
$\text{LR} = 5 \times 10^{-5} \times \sqrt{10}$	0.257
$\text{LR} = 5 \times 10^{-4}$	0.124
$\text{LR} = 5 \times 10^{-5} \times \sqrt{10}$ Regularized	0.385
50-Assay Batch	0.415

Table A3. The complete results (ρ_{assay}) of aggregate correlations across all ChEMBL data sets and folds. ρ_{assay} for DTI trained on omnivore_short was not available at the time of submission.

dataset	fold	direct	ranking (inter-assay)	ranking (intra-assay)
kinodata	1	0.324	0.363	0.386
	2	0.325	0.357	0.399
	3	0.286	0.409	0.385
	4	0.324	0.289	0.316
	5	0.352	0.366	0.215
landrum	1	0.292	0.253	0.399
	2	0.262	0.312	0.384
	3	0.237	0.245	0.393
	4	0.299	0.259	0.381
	5	0.258	0.267	0.397
omnivore	1	0.373	0.387	0.398
	2	0.359	0.351	0.383
	3	0.351	0.354	0.376
	4	0.348	0.375	0.367
	5	0.377	0.388	0.372
omnivore_short	1	-	0.349	0.402
	2	-	0.325	0.386
	3	-	0.348	0.405
	4	-	0.318	0.372
	5	-	0.333	0.421